

- 3 (a) Enzymes are polymers of amino acids.

Complete Fig. 3.1 to show the general structure of an amino acid.

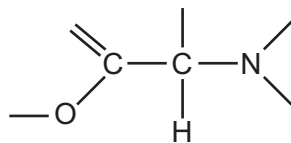


Fig. 3.1

[1]

- (b) When bananas are peeled, the exposed tissue gradually turns brown in the presence of oxygen in the air. This is due to an enzyme called catechol oxidase, which acts on the substrate catechol. Catechol and catechol oxidase are present in the banana tissue.

The overall reaction is shown in Fig. 3.2.

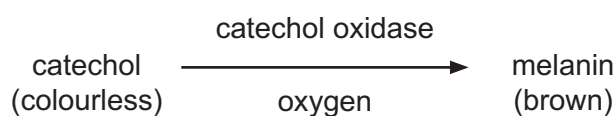


Fig. 3.2

A student investigated how the concentration of catechol oxidase affects the rate of this reaction. All other variables were kept constant throughout the investigation.

For each concentration of catechol oxidase used, the student mixed catechol oxidase solution with catechol and recorded the time taken for the mixture to reach a standard brown colour.

The rate of reaction, **R**, for each concentration of catechol oxidase used was then calculated using the formula:

$$R = \frac{1}{\text{time to reach standard brown colour in minutes}}$$

- (i) Calculate the rate of reaction when the standard brown colour was reached in 2 minutes 30 seconds.

rate of reaction = min⁻¹ [1]

(ii) Fig. 3.3 is a graph showing the results of the investigation.

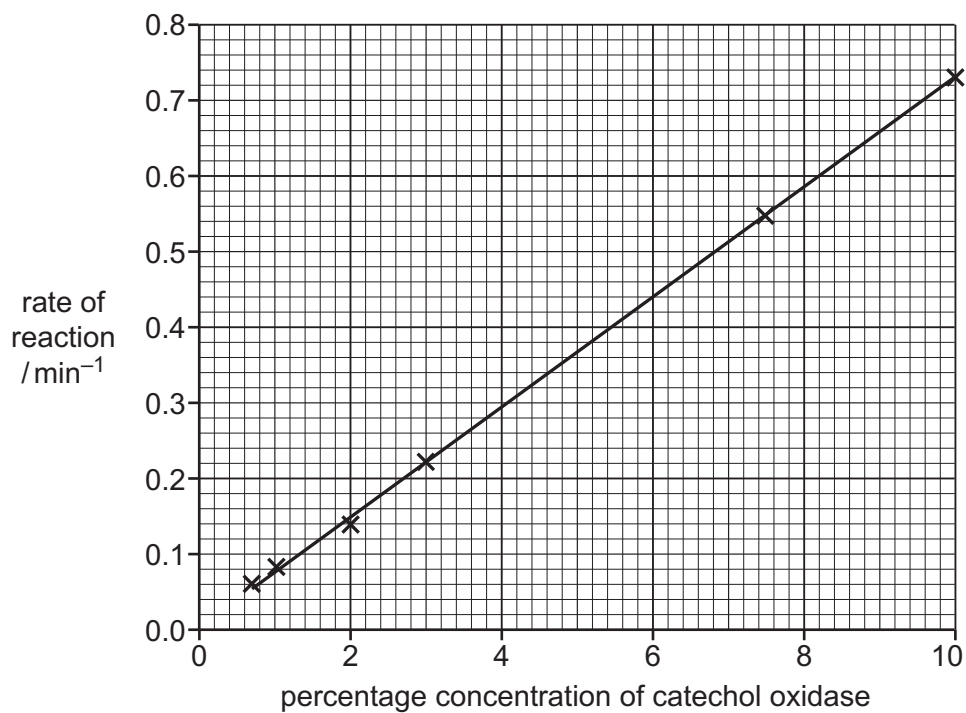


Fig. 3.3

State how the results shown in Fig. 3.3 show that substrate was in excess at all concentrations of catechol oxidase tested.

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..... [1]

- (c) The student carried out a further experiment to investigate how the concentration of catechol affects the initial rate of reaction. All other variables were kept constant throughout this investigation.

Fig. 3.4 is a graph showing the effect of varying the concentration of catechol on the initial rate of reaction.

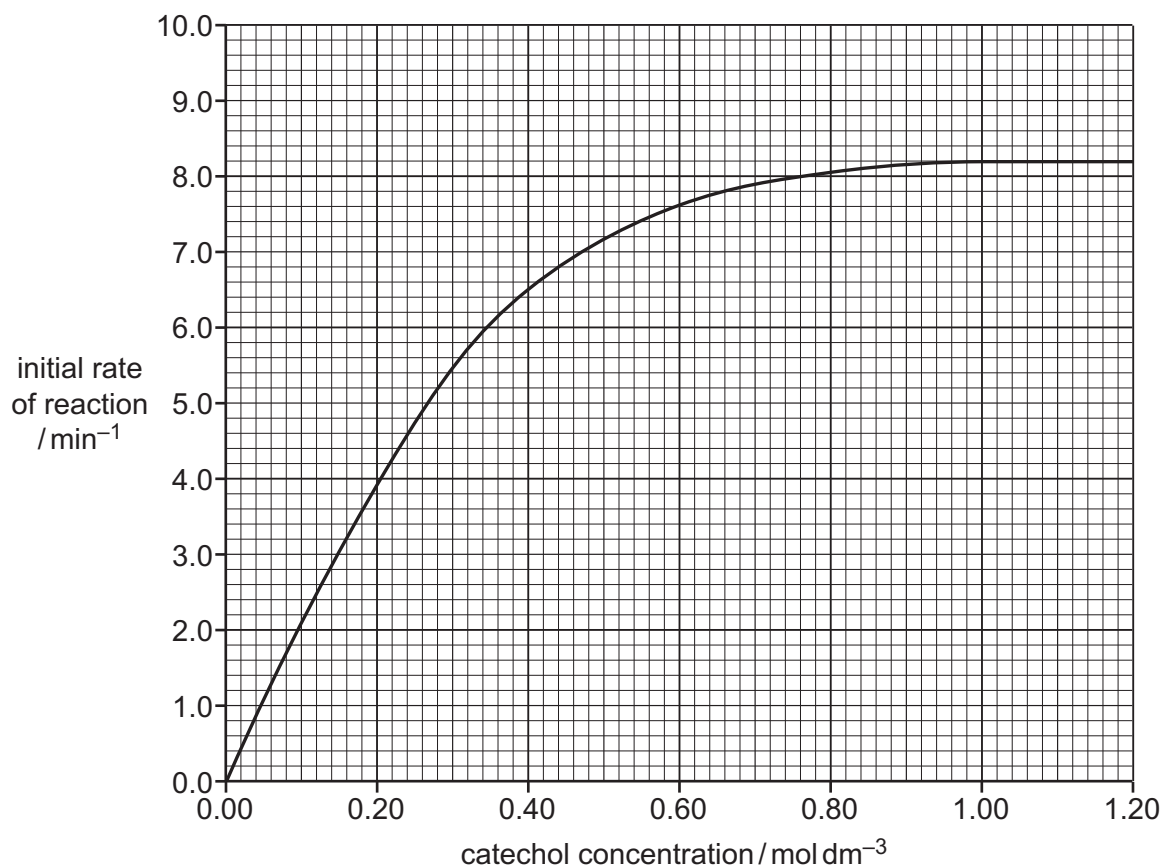


Fig. 3.4

- (i) Explain the shape of the curve shown in Fig. 3.4.

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- (ii) Use Fig. 3.4 to calculate the value of the Michaelis–Menten constant (K_m) for the reaction between catechol oxidase and catechol.

$$K_m = \dots\dots\dots \text{mol dm}^{-3} \quad [1]$$

- (iii) Methylcatechol has a similar shape to catechol. Catechol oxidase can also use methylcatechol as a substrate.

The K_m value for the reaction using methylcatechol as the substrate was found to be much lower than the K_m value for the reaction using catechol as the substrate, when the reactions were carried out under the same conditions.

State what these K_m values indicate about the relationship between the enzyme and the two substrates.

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..... [1]

[Total: 8]